SCIENCE PROGRAM 2015

STRAND: Energy and Change TOPIC: Motion YEAR: 10 Extension TERM: Autumn (8 weeks)

TIME (WEEK)	STUDENT OUTCOMES	PRACTICALS	TEXT REFERENCE	ASSESSMENT/ HOMEWORK
1-2	**Speed and Velocity** List common examples of scalar and vector quantities. Understand the difference between a scalar and vector quantity. Understand the difference between distance and displacement. Be able to determine displacement by construction and using scales. **Extension** - Be able to calculate displacement by using Pythagoras' Theory and trigonometry. Understand the difference between speed and velocity. Be able to convert kmh^{-1} to ms^{-1}. Be able to measure motion using a ticker timer. Apply the equations: speed = d/t., v = s/t. **Extension** - Be able to graph and represent motion - distance-time, displacement-time, speed-time, velocity-time. Be able to calculate displacement from the area under a velocity-time graph.	Activity: using maps to work out distance and displacement. Invest. 8.1 - ticker timer	Ch. 8.1 SQ10 Ch. 8.2 SQ10	Worksheet 8.1 - Speed + velocity Worksheet 8.2 - Ticker timers
3	**Acceleration** Be able to calculate the acceleration of an object using a = (v-u)/t. Understand that the acceleration due to gravity near the Earth's surface is 9.80 ms^{-2}. **Extension** - Apply the equations of motion ($v = u + at$, $s = ut + {}^1/_2at^2$, $v^2 = u^2 + 2as$). Be able to draw graphs to represent motion - acceleration-time. Be able to calculate the gradient of a graph and understand what it represents.	Invest. 8.2 - acceleration Expt 1.3 (BI 10) - reaction time	Ch. 8.3 SQ10	Worksheet 8.3 - Acceleration

4-5	**Newton's Laws of Motion** First Law - understand the concept of inertia. Second Law - understand that an unbalanced force will cause an object to accelerate. Understand the difference between mass and weight. Be able to perform calculations using $F = ma$ and $F_w = mg$. Third Law - understand the concept of action and reaction.	Invest. 8.4 - Forces, mass + acceleration Invest. 8.6 - Balloon rocket	Ch. 8.4 SQ10 Ch. 8.5 SQ10 Ch. 8.6 SQ10	Worksheet 8.4 - Inertia + motion Worksheet 8.5 - Force and gravity Worksheet 8.6 - Newton's Second Law Worksheet 8.7 - Newton's Third Law **Assignment - Motion**
6-7	**Energy and Work** Understand the connection between work and energy. Be able to calculate work using $W = Fs$. Be able to define gravitational potential energy (E_p) and kinetic energy (E_k). **Extension** - Be able to perform calculations using $E_p = mgh$ and $E_k = {}^1/_2\, mv^2$. Understand and apply the Law of Conservation of Energy. Understand that no process is 100% efficient.	Invest. 8.7 - Bouncing ball	Ch. 8.7 SQ10 Ch. 8.8 SQ10	**Test - Physical Sciences**

Texts Science Quest 10 (SQ10)
Big Ideas 10 (BI10)